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ASSIGNMENT 1

6.2, 6.3, 6.7, 6.8, 6.9, 6.10, 6.11 (English texthook)

6.2 (SR latch) Draw the output timing diagram of (a) NOR and (b) NAND implementation of an SR latch for the input signals depicted in Figure P6.2.

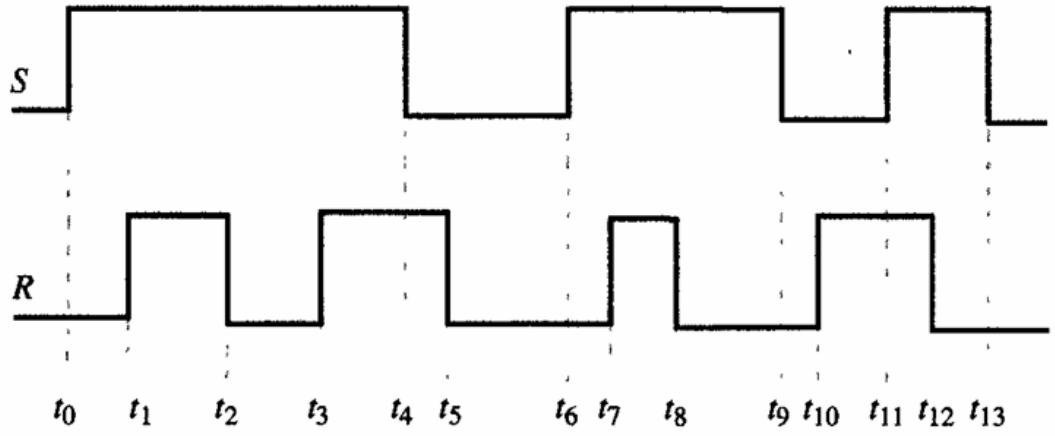
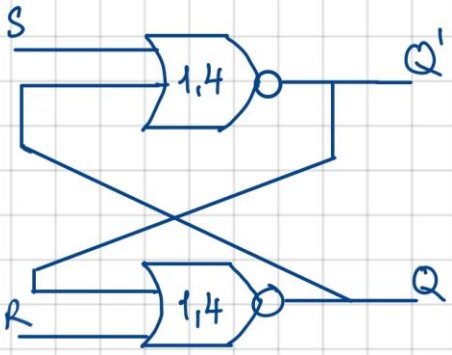


FIGURE P6.2

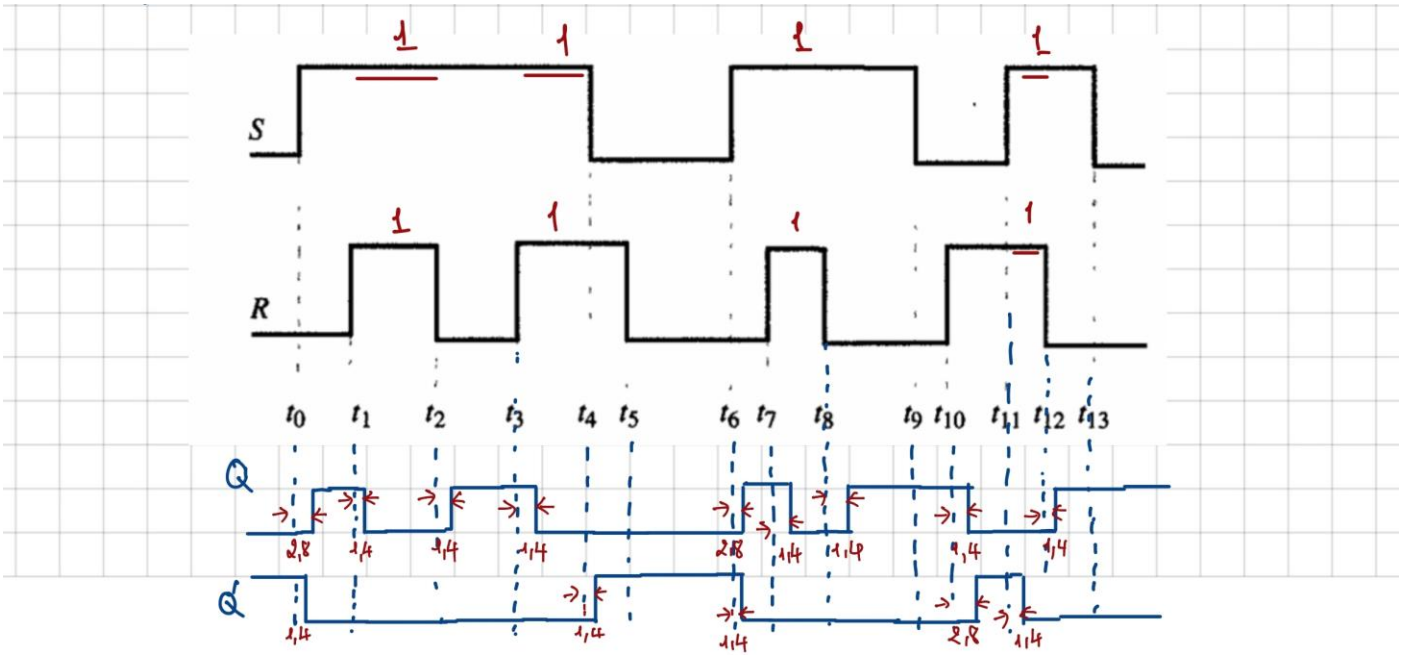
Solution:

a) SR latch (NOR implementation)

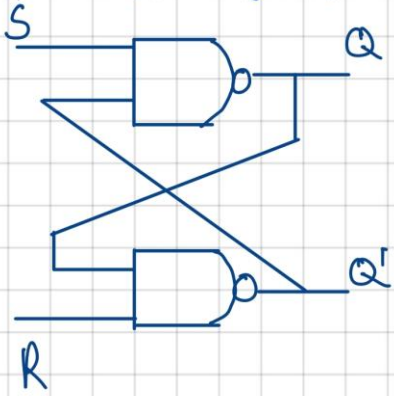


Truth Table:

S	R	Q(next)	Q'(next)	
0	0	Qlast	Q'last	(hold)
0	1	0	1	(reset)
1	0	1	0	(set)
1	1	0	0	(? undefined)



b) SR latch (NAND implementation)



S	R	Q_{next}	Q'_{next}	
0	0	1	1	(? undefined)
0	1	1	0	(set)
1	0	0	1	(reset)
1	1	Q_{last}	Q'_{last}	(hold)

6.3 (SR latch) Derive an implementation of a clocked SR latch using only:

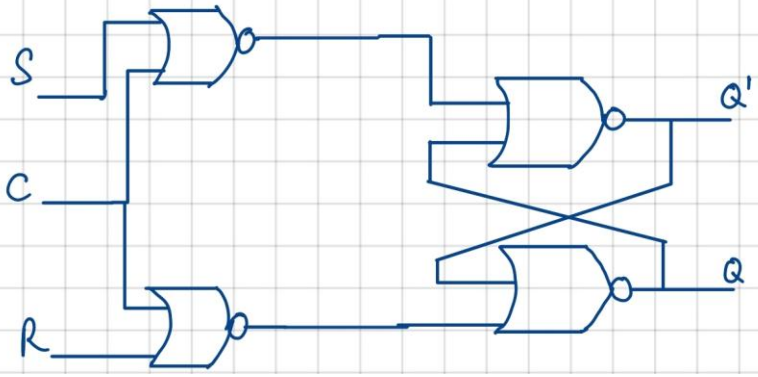
(a) NOR gates

(b) NAND gates

(c) AND, OR, and INVERT gates

Solution:

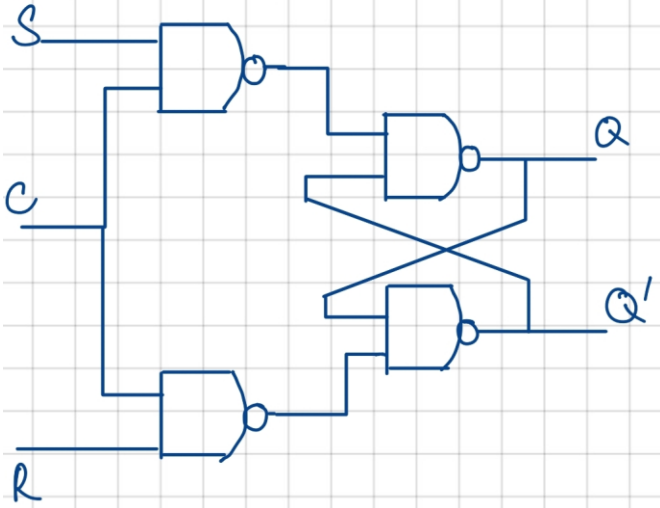
a) NOR gates



Truth Table:

C	S	R	Q_{next}	Q'_{next}	
1	X	X	Q_{last}	Q'_{last}	(hold)
0	0	0	0	0	(hold)
0	0	1	1	0	(set)
0	1	0	0	1	(reset)
0	1	1	Q_{last}	Q'_{last}	(hold)

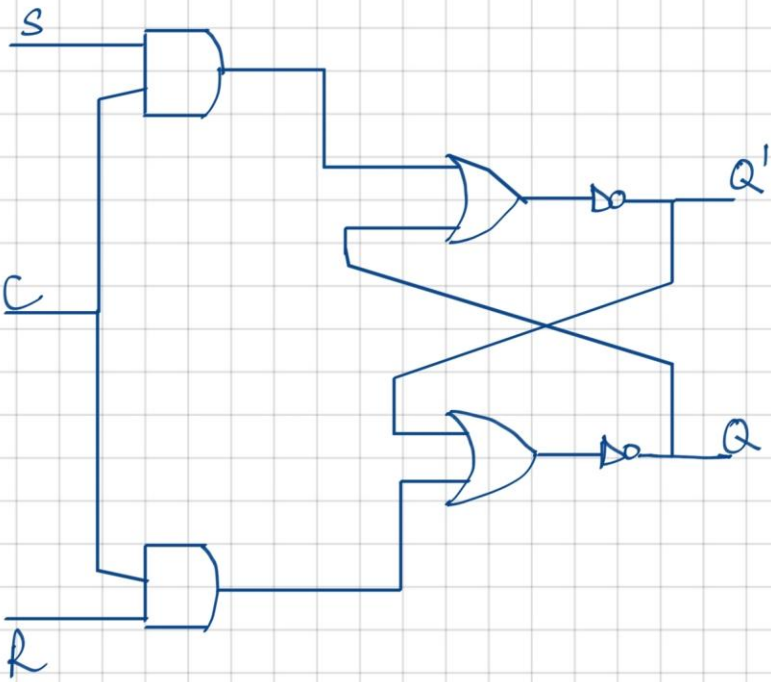
b) NAND gates



Truth Table:

C	S	R	Q_{next}	Q'_{next}	
0	X	X	Q_{last}	Q'_{last}	(hold)
1	0	0	Q_{last}	Q'_{last}	(hold)
1	0	1	0	1	(reset)
1	1	0	1	0	(set)
1	1	1	1	1	(?)

c)



Truth Table:

C	S	R	Q_{next}	Q'_{next}	
0	X	X	Q_{last}	Q'_{last}	(hold)
1	0	0	Q_{last}	Q'_{last}	(hold)
1	0	1	0	1	(reset)
1	1	0	1	0	(set)
1	1	1	0	0	(? undefined)

6.7 (Flip-flops) Using Karnaugh maps, derive the characteristic equations from the characteristic tables of the four flip-flops shown in Table 6.1.

a) SR - FF

Characteristic Table:

S	R	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	NA
1	1	1	NA

K - map:

Q^+ / SR \ Q	0	1
00	0	1
01	0	0
11	X	X
10	1	1

Characteristic equations:

$$Q^+ = S + R'Q$$

b) JK - FF

Characteristic Table:

J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

K - map:

Q^+ / JK \ Q	0	1
00	0	1
01	0	0
11	1	0
10	1	1

Characteristic equations:

$$Q^+ = JQ' + K'Q$$

c) D - FF

Characteristic Table:

D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

K - map:

Q^+ / D \ Q	0	1
0	0	0
1	1	1

Characteristic equations:

$$Q^+ = D$$

d) T-FF

Characteristic Table:

T	Q	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

K-map:

T \ Q^+ Q	0	1
0	0	1
1	1	0

Characteristic equation:

$$Q^+ = TQ' + T'Q$$

6.8 (Flip-flops) Implement SR, JK, D, and T flip-flops using only AND, OR, and INVERT gates, and:

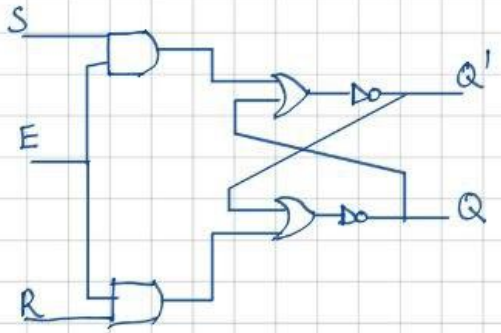
- **(a)** SR flip-flops
- **(b)** JK flip-flops
- **(c)** D flip-flops
- **(d)** T flip-flops

Solutions:

6.8

a) SR Flip-Flops

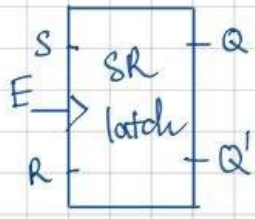
* Implement clocked SR latch using OR, NOR, INVERT gates:



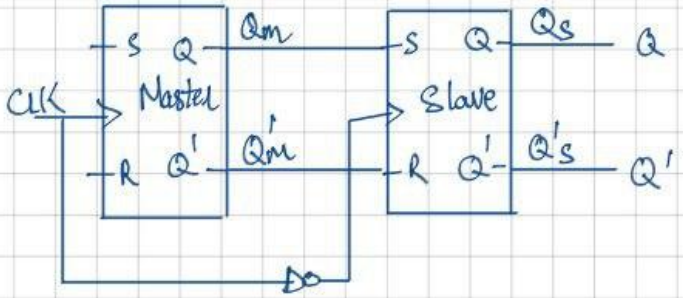
Truth Table:

C	S	R	Q_{next}	Q'_{next}	
0	X	X	Q_{last}	Q'_{last}	(hold)
1	0	0	Q_{last}	Q'_{last}	(hold)
1	0	1	0	1	(reset)
1	1	0	1	0	(set)
1	1	1	0	0	(? undefd)

Packaging this latch:

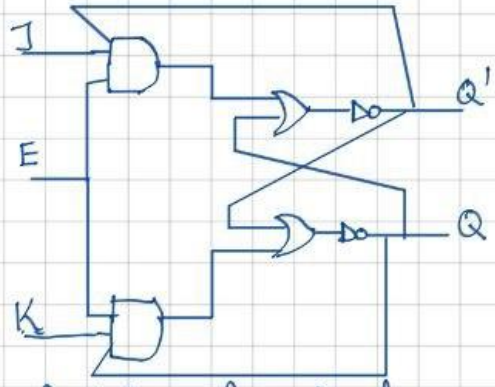


* Implement master-slave SR Flip-Flop:



b) JK Flip-Flops

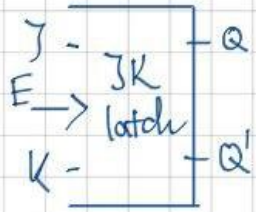
* Implement clocked JK latch using OR, NOR, INVERT gates:



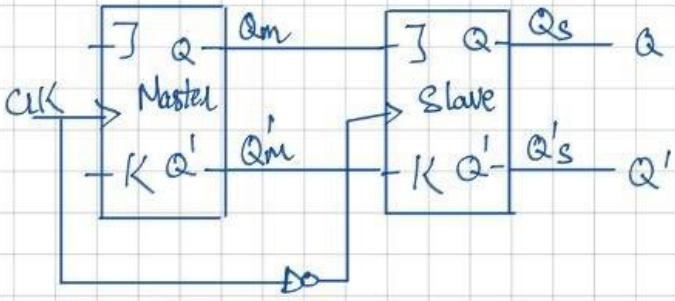
Truth Table:

C	J	K	Q_{next}	Q'_{next}	
0	X	X	Q_{last}	Q'_{last}	(hold)
1	0	0	Q_{last}	Q'_{last}	(hold)
1	0	1	0	1	(reset)
1	1	0	1	0	(set)
1	1	1	Q_{last}	Q_{last}	(toggle)

Packing this latch:

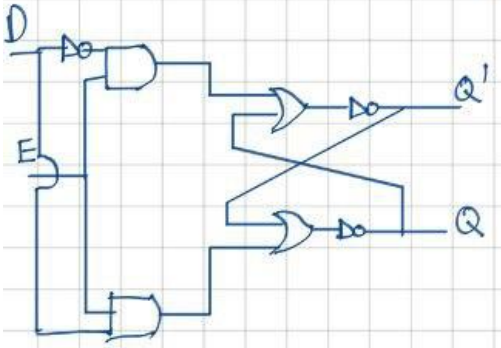


* Implement master-slave JK Flip-Flop:



c) D Flip-Flops

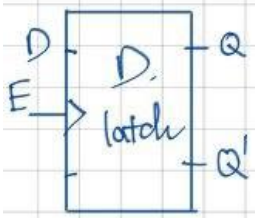
* Implement clocked D latch using OR, NOR, INVERT gates:



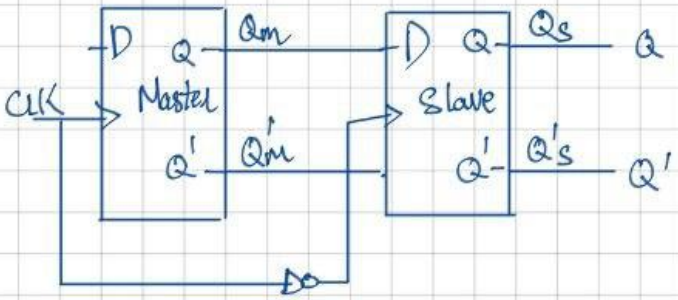
Truth Table:

C	D	Q_{next}	Q'_{next}	
0	X	Q_{last}	Q'_{last}	(hold)
1	0	0	1	(reset)
1	1	1	0	(set)

Packing this circuit:

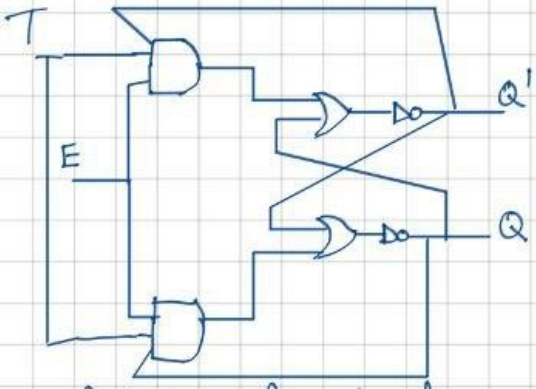


* Implement master-slave D Flip Flop:



d) T Flip-Flops

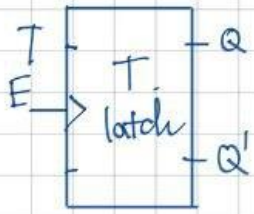
* Implement clocked JK latch using OR, NOR, INVERT gates:



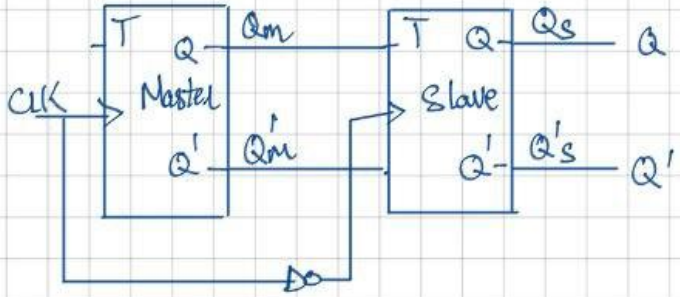
Truth Table:

C	T	Q_{next}	Q'_{next}	
0	X	Q_{last}	Q'_{last}	(hold)
1	0	Q_{last}	Q'_{last}	(hold)
1	1	Q'_{last}	Q_{last}	(toggle)

Packing this latch:



* Implement master-slave T Flip-Flop:



6.9 (Sequential analysis) Derive a (a) state table and (b) state diagram for the sequential circuit shown in Figure P6.9.

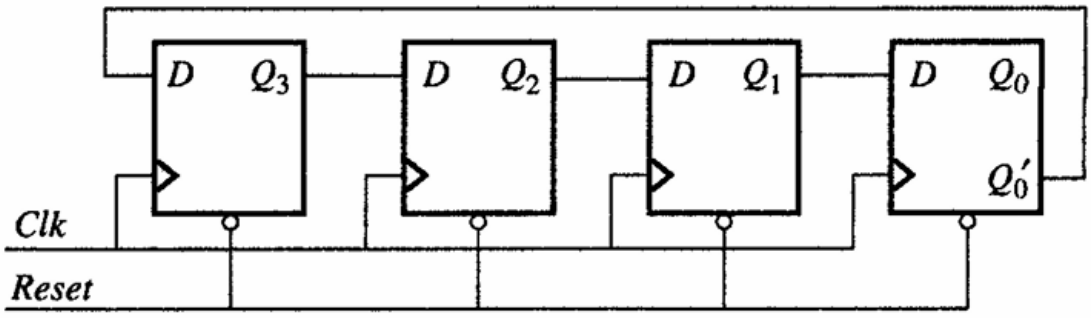


FIGURE P6.9

Solutions:

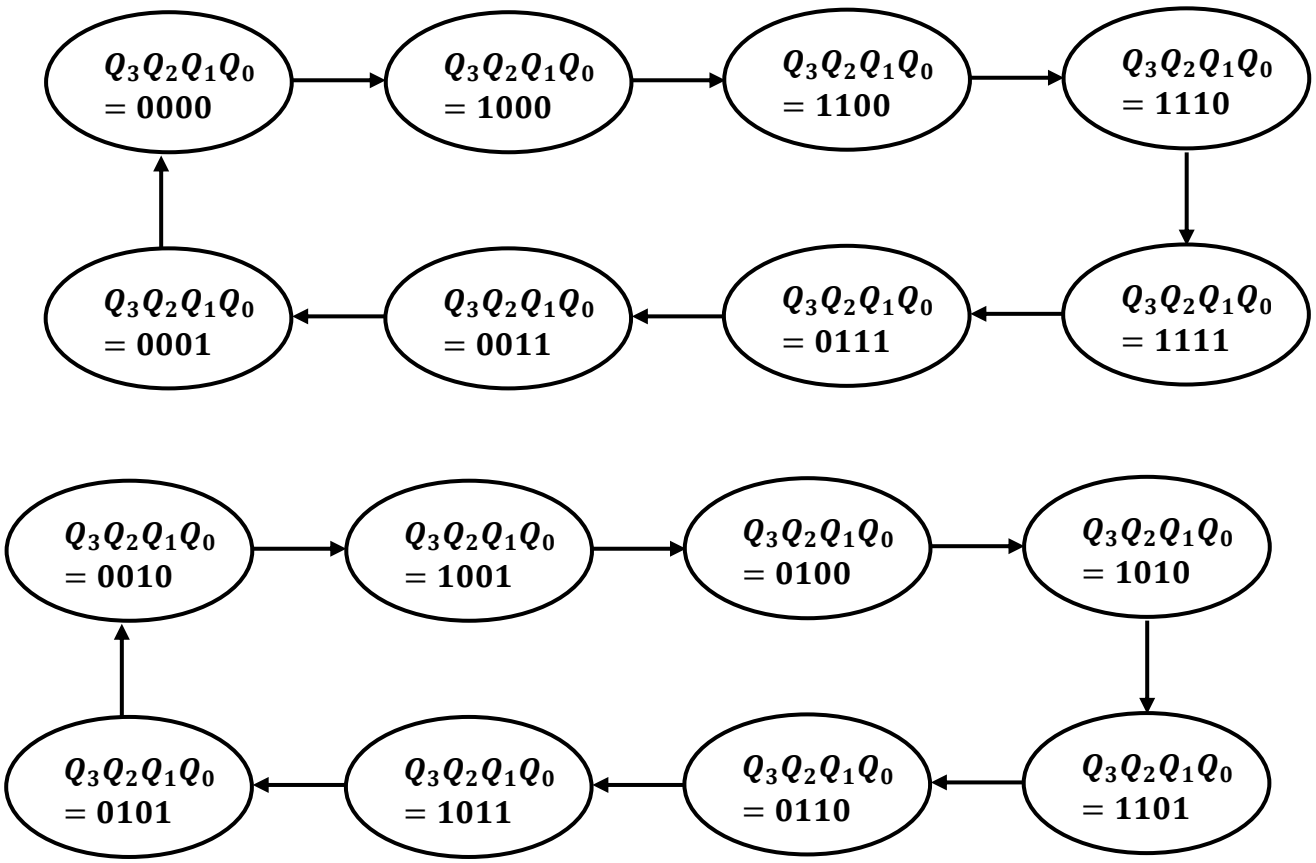
a. Excitation Equations:

$$D_3 = Q'_0$$
$$D_2 = Q_3$$
$$D_1 = Q_2$$
$$D_0 = Q_1$$

State Table:

Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	1	0	0	0	1	0	0	0
0	0	0	1	0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1	1	0	0	1
0	0	1	1	0	0	0	1	0	0	0	1
0	1	0	0	1	0	1	0	1	0	1	0
0	1	0	1	0	0	1	0	0	0	1	0
0	1	1	0	1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	1	0	0	1	1
1	0	0	0	1	1	0	0	1	1	0	0
1	0	0	1	0	1	0	0	0	1	0	0
1	0	1	0	1	1	0	1	1	1	0	1
1	0	1	1	0	1	0	1	0	1	0	1
1	1	0	0	1	1	1	0	1	1	1	0
1	1	0	1	0	1	1	0	0	1	1	0
1	1	1	0	1	1	1	1	1	1	1	1
1	1	1	1	0	1	1	1	0	1	1	1

b. State diagram



6.10 (Sequential analysis) Derive the (a) state/output table and (b) FSM representation of the circuit shown in Figure P6.10. What is the function of this sequential circuit?

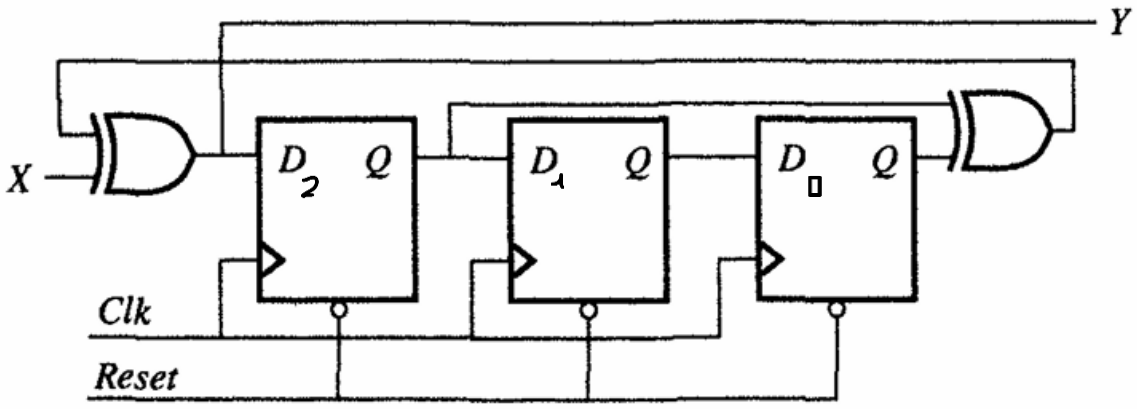


FIGURE P6.10

Solutions:

a. Excitation Equations:

$$Y = X \oplus Q_0 \oplus Q_2$$
$$D_2 = Q_0 \oplus Q_2$$
$$D_1 = Q_2$$
$$D_0 = Q_1$$

State Table:

Reset	Q_2	Q_1	Q_0	D_2	D_1	D_0	Q_2^+	Q_1^+	Q_0^+	**Y**	
										X = 0	**X =1**
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	1	0	0	1	0	0	1	0
1	0	1	0	0	0	1	0	0	1	0	1
1	0	1	1	1	0	1	1	0	1	1	0
1	1	0	0	1	1	0	1	1	0	1	0
1	1	0	1	0	1	0	0	1	0	0	1
1	1	1	0	1	1	1	1	1	1	1	0
1	1	1	1	0	1	1	0	1	1	0	1
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	1	0	0	1	0	0	1	0
1	0	1	0	0	0	1	0	0	1	0	1
1	0	1	1	1	0	1	1	0	1	1	0
1	1	0	0	1	1	0	1	1	0	1	0
1	1	0	1	0	1	0	0	1	0	0	1
1	1	1	0	1	1	1	1	1	1	1	0
1	1	1	1	0	1	1	0	1	1	0	1

6.11 (Sequential analysis) Derive (a) excitation equations, (b) a next-state equation, (c) a state/output table, and (d) a state diagram for the circuit shown in Figure P6.11.

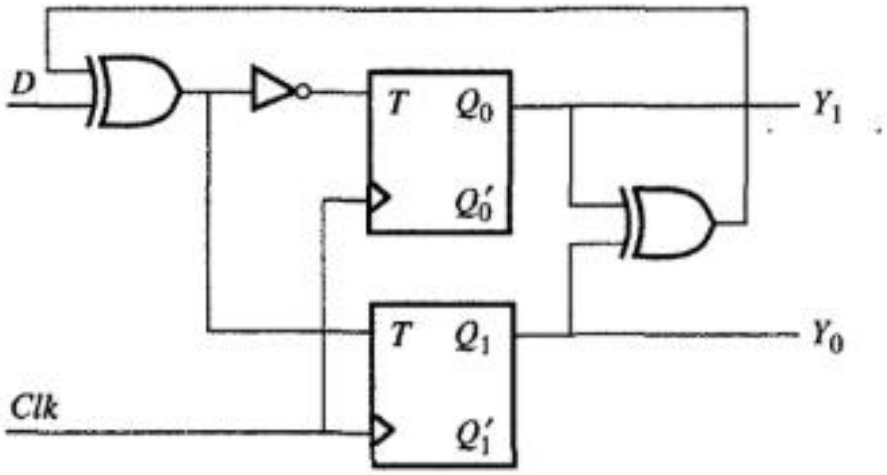


FIGURE P6.11

Solution:

a. Excitation equations:

$$T_0 = \overline{D \oplus (\ Q_0 \oplus Q_1)}$$

$$T_1 = D \oplus (\ Q_0 \oplus Q_1) = \overline{T_0}$$

$$Y_1 = Q_0$$

$$Y_0 = Q_1$$

b. Next-State equation:

$$Q_0^+ = T'_0 Q_0 + T_0 Q'_0$$

$$Q_1^+ = T'_1 Q_1 + T_1 Q'_1$$

c. State/Output Table:

D	Q_1	Q_0	T_1	T_0	Q_1^+	Q_0^+	$Y_0 = Q_1$	$Y_1 = Q_0$
0	0	0	0	1	0	1	0	0
0	1	0	1	0	0	0	1	0
0	0	1	1	0	1	1	0	1
0	1	1	0	1	1	0	1	1
1	0	0	1	0	1	0	0	0
1	1	0	0	1	1	1	1	0
1	0	1	0	1	0	0	0	1
1	1	1	1	0	0	1	1	1